



**BAULKHAM HILLS
HIGH
SCHOOL**

2018

**YEAR 12
TERM 2
ASSESSMENTS**

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Show all relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work
- A reference sheet is provided at the back of this paper

**Total marks:
41**

Attempt all Questions 1 – 4 (pages 2-5)

Question 1 (9 Marks) Start on the appropriate page of your answer booklet.

a) Find the value of the constant term in the expansion of $\left(3x^2 + \frac{2}{x}\right)^{12}$. 2

b) The motion of an object is defined by the equation $x = 4\sin(\pi t)$ where x is the displacement from the origin in centimetres at time t seconds.

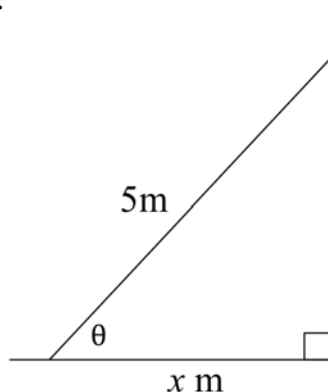
(i) Write down the amplitude of the motion. 1

(ii) What is the period of the motion in seconds? 1

(iii) What is the greatest speed of the object? 1

(iv) Prove that the object is moving in Simple Harmonic Motion. 1

c) A 5 metre ladder rests with one end against a wall and the other end on horizontal ground which is level with the base of the wall and x metres from the bottom of the wall. The end which is in contact with the ground is sliding away from the wall at a constant rate of 0.1 m/s.



Find the rate (in radians/s) at which the angle, θ , between the ladder and the ground 3 is decreasing when the end of the ladder is 3 metres from the wall.

End of Question 1

Question 2 (13 Marks) Start on the appropriate page of your answer booklet.

- a) The acceleration of a particle undergoing Simple Harmonic Motion is given by $\ddot{x} = -n^2x$, where x is the displacement of the particle from the origin in metres, and n is a positive constant. The particle starts from rest at a distance of 10 metres to the right of its centre of oscillation O . The period of the motion is 2 seconds.
- (i) Prove that $v^2 = \pi^2(100 - x^2)$, where v is the velocity of the particle in metres per second. **3**
- (ii) Find the speed of the particle in metres per second when it is 6 metres from the origin. **1**
- (iii) Find the time taken by the particle to first reach the point 6 metres to the right of O , correct to 2 decimal places. (You may assume $x = a \cos nt$ where a is a constant is a solution of $\ddot{x} = -n^2x$). **2**
- b) The rate of growth of a population N over t years is given by: $\frac{dN}{dt} = -k(N - 700)$.
- (i) Show that $N = 700 + Ae^{-kt}$ satisfies $\frac{dN}{dt} = -k(N - 700)$ where A and k are constants. **1**
- (ii) The population decreased from an initial population of 8300 to 5100 in 5 years. Find the population at the end of the next 5 years. Give your answer correct to the nearest hundred. **3**
- c) Consider the binomial expansion $1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1 + x)^n$.
- (i) Show that $1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$. **1**
- (ii) Show that
$$\left[1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}\right] \left[1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n}\right] = 3^n.$$
 2

End of Question 2

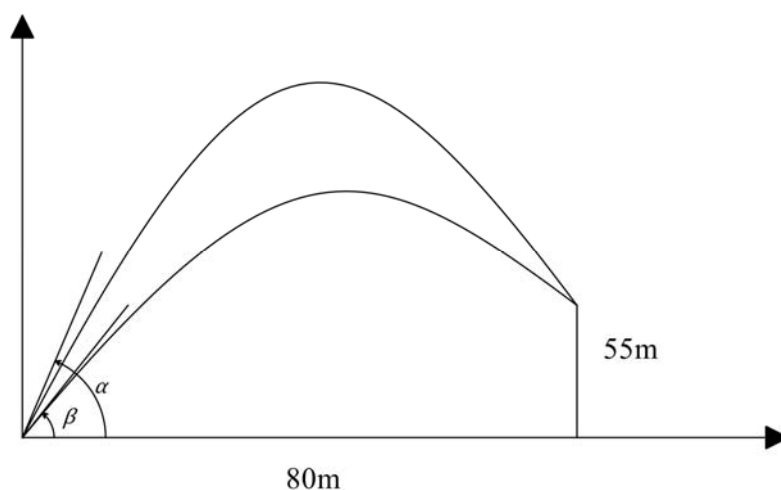
Question 3 (9 Marks) Start on the appropriate page of your answer booklet.

- a) The coefficients of x^4 and x^5 in the expansion of $(2 + 5x)^n$ are in the ratio of 1:3 2
Find the value of n .
- b) A certain particle moves along the x axis according to the equation $t = 2x^2 - 5x + 3$,
where x is its displacement measured in centimetres at time t seconds.
Initially the particle is 1.5 cm to the right of the origin, O , and moving away
from O .
- (i) Prove that the velocity, $v \text{ cm s}^{-1}$, is given by $v = \frac{1}{4x - 5}$. 2
- (ii) Find the velocity of the particle when $t = 6$ seconds. 3
- (iii) Find an expression for the acceleration, $a \text{ cm s}^{-2}$, in terms of x . 2

End of Question 3

Question 4 (10 Marks) Start on the appropriate page of your answer booklet.

- a) A missile is fired from ground level into the air at a velocity of 40m/s and at an angle of α with the horizontal.



A short time later another missile is fired from the same point and with the same speed but at a different angle β . Both missiles hit the same target at the same time. The target is 55m above the ground and 80m horizontally from the point of firing. (Take $g = 10\text{m/s}^2$ and neglect air resistance).

Given the equations of motion at time t seconds are:

$$x = 40t\cos\alpha \text{ and } y = -5t^2 + 40t\sin\alpha \text{ (DO NOT PROVE)}$$

- (i) Show that the path of the first missile is given by $y = x\tan\alpha - x^2\left(\frac{\sec^2\alpha}{320}\right)$. 2
- (ii) Find the values of $\tan\alpha$ and $\tan\beta$. 2
- (iii) Determine the time difference between the firing times of the two missiles. 2

- b) Consider the expansion $(1-x)^n = \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \dots + (-1)^n\binom{n}{n}x^n$.

- (i) Show that $\int_0^1 (1-x)^n dx = \binom{n}{0} - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + \frac{(-1)^n}{n+1}\binom{n}{n}$. 2

- (ii) Hence find the value of $\sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r}$. 2

End of Paper

HSC EXTENSION 1 TASK 3 JUNE 2018

Q1 a)

$$\begin{aligned}\text{General term: } & {}^{12}C_k (3k)^{12-k} (2k)^k \\ &= {}^{12}C_k 3^{12-k} 2^k k^{24-2k} \\ &= {}^{12}C_k 3^{12-k} 2^k k^{24-3k}\end{aligned}$$

(2) correct answer

(1) finds general term
NB unsimplified is acceptable

For constant term $24-3k=0$

$$k=8$$

$$\therefore \text{Constant term is } {}^{12}C_8 3^4 2^8 (=10264320)$$

b (i) Amplitude = 4

(1) correct answer

$$\begin{aligned}\text{(ii) Period} &= \frac{2\pi}{\pi} \\ &= 2\end{aligned}$$

(1) correct answer

$$\text{(iii) } x = 4\pi \cos(\pi t)$$

(1) correct answer

$\therefore$ Greatest speed = 4π cm/s.

$$\text{(iv) } \ddot{x} = -4\pi^2 \sin(\pi t)$$

(1) correct solution

$$\ddot{x} = -\pi^2 x \text{ which is of the form } \ddot{x} = -\omega^2 x$$

$\therefore$ Object is moving in SHM.

$$\text{c) } \cos \theta = \frac{x}{5}$$

$$\theta = \cos^{-1} \frac{x}{5}$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}$$

$$\frac{d\theta}{dt} = \frac{-1}{\sqrt{25-x^2}} \times 0.1$$

$$\text{When } x=3 \quad \frac{d\theta}{dt} = \frac{-1}{\sqrt{16}} \times \frac{1}{10}$$

$$= -\frac{1}{40} \text{ radians/sec.}$$

(3) correct solution

(2) finds $\frac{d\theta}{dx}$ or $\frac{dx}{d\theta}$

(1) establishes correct chain rule

$$2 \text{ a) (i) Period} = \frac{2\pi}{n}$$

$$2 = \frac{2\pi}{n}$$

$$n = \pi$$

$$\ddot{x} = -\pi^2 x$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -\pi^2 x$$

$$\left[\frac{1}{2} v^2 \right]_0^v = -\pi^2 \left[\frac{x^2}{2} \right]_0^x$$

$$\frac{1}{2} (v^2 - 0) = -\pi^2 \left(\frac{x^2}{2} - \frac{10^2}{2} \right)$$

$$\frac{v^2}{2} = \pi^2 \left(\frac{100 - x^2}{2} \right)$$

$$v^2 = \pi^2 (100 - x^2)$$

③ correct solution

② finds $\frac{1}{2} v^2$ in terms of x (including constant)

① finds the value of n
OR ① attempts to integrate
 $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -n^2 x$

$$(ii) \text{ When } x=4, v^2 = \pi^2 (100 - 36)$$

$$v^2 = 64\pi^2$$

$$v = \pm \pi \sqrt{64}$$

$$\therefore \text{Speed} = 8\pi \text{ m/s}$$

① correct answer

$$(iii) \text{ When } x=6, 6 = 10 \cos \pi t$$

$$\frac{3}{5} = \cos \pi t$$

$$\pi t = 0.92729 \dots$$

$$\therefore t = 0.295167 \dots$$

$$t = 0.30 \text{ seconds (2dp)}$$

② correct answer

① substitutes $x=6$ into

$$x = 10 \cos \pi t$$

2(b)(i)

$$N = 700 + Ae^{-kt}$$

$$\frac{dN}{dt} = -Ake^{-kt}$$

$$= -k(700 + Ae^{-kt} - 700)$$

$$= -k(N - 700)$$

① correct solution

ii) When $t=0$, $N=8300$

$$8300 = 700 + Ae^0$$

$$A = 7600$$

$$\therefore N = 700 + 7600e^{-kt}$$

When $t=5$, $N=5100$

$$5100 = 700 + 7600e^{-5k}$$

$$4400 = 7600e^{-5k}$$

$$\frac{11}{19} = e^{-5k}$$

$$-5k = \ln \frac{11}{19}$$

$$5k = \ln \frac{19}{11}$$

$$k = \frac{\ln \frac{19}{11}}{5}$$

$$(\approx 0.1093...)$$

③ correct answer

② finds A and k

① finds correct value of A and attempts to find k

$$\text{When } t=10, N = 700 + 7600e^{-10 \times \frac{\ln \frac{19}{11}}{5}}$$

$$= 3247.368...$$

$\therefore$ Population is approximately 3200

2(c) i) Let $x=1$

$$1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = (1+1)^n$$

$$1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

① correct solution

$$\text{ii) let } x = \frac{1}{2} \text{ in } 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n \quad \text{② correct solution}$$

$$\therefore 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n} = (1 + \frac{1}{2})^n \quad \text{① substitutes } x = \frac{1}{2}$$

$$\left[1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}\right] \left[1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n}\right] = 2^n \times 1.5^n = (2 \times 1.5)^n$$

$$= 3^n$$

3 (a) General term: ${}^nC_k 2^{n-k} (5x)^k$

$$\therefore \frac{{}^nC_4 2^{n-4} 5^4}{{}^nC_5 2^{n-5} 5^5} = \frac{1}{3}$$

$$\frac{2}{5} \frac{\frac{n!}{(n-4)! 4!}}{n!} = \frac{1}{3}$$

$$\frac{(n-5)! 5!}{(n-4)(n-5)!} \times \frac{8}{6} = \frac{8}{6}$$

$$n-4=6$$

$$n=10$$

② correct answer

① equates ratio of coefficients to $\frac{1}{3}$.

b) i) $t = 2x^2 - 5x + 3$

$$\frac{dt}{dx} = 4x - 5$$

$$\frac{dx}{dt} = \frac{1}{4x-5}$$

② correct solution

① finds $\frac{dt}{dx}$

(ii) When $t=6$, $6 = 2x^2 - 5x + 3$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$x = -\frac{1}{2}, x = 3$$

But when $t=0$, $x = 1.5$ and $v = \frac{1}{1} = 1 \text{ cm/s}$

Since $v \neq 0$, particle never stops and can't reach $\frac{1}{2}$.

$$\therefore v = \frac{1}{4x-5} = \frac{1}{-7} \text{ cm/s}$$

③ correct solution

② finds $v = \pm \frac{1}{7}$

① calculates $x=3$

iii) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$= \frac{1}{2} \frac{d(4x-5)^{-2}}{dx}$$

$$= \frac{1}{2} \times -2 \times 4 (4x-5)^{-3}$$

$$a = \frac{-4}{(4x-5)^3}$$

② correct solution

① attempts to differentiate

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

4 a)(i) From $x = 40t \cos \alpha$
 $t = \frac{x}{40 \cos \alpha}$

sub in $y = -5t^2 + 40t \sin \alpha$

$$y = -5 \frac{x^2}{1600 \cos^2 \alpha} + \frac{40}{40 \cos \alpha} x \sin \alpha$$

$$y = x \tan \alpha - \frac{x^2}{320} \sec^2 \alpha$$

(ii) When $x = 80$ $y = 55$

$$55 = 80 \tan \alpha - \frac{6400 \sec^2 \alpha}{320}$$

$$55 = 80 \tan \alpha - 20(\tan^2 \alpha + 1)$$

$$20 \tan^2 \alpha - 80 \tan \alpha + 175 = 0$$

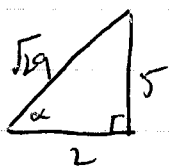
$$4 \tan^2 \alpha - 16 \tan \alpha + 115 = 0$$

$$(2 \tan \alpha - 3)(2 \tan \alpha - 5) = 0$$

$$\tan \alpha = \frac{3}{2}, \tan \alpha = \frac{5}{2}$$

$$\therefore \tan \alpha = \frac{5}{2} \text{ \& } \tan \beta = \frac{3}{2}$$

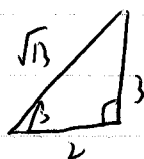
iii)



$$x = 40t \cos \alpha$$

$$80 = 40t \frac{2}{\sqrt{29}}$$

$$t = \sqrt{29}$$



$$x = 40t \cos \beta$$

$$80 = 40t \frac{2}{\sqrt{13}}$$

$$t = \sqrt{13}$$

$$\therefore \text{Difference in time} = \sqrt{29} - \sqrt{13}$$

$$= 1.7796 \dots \text{ sec}$$

(2) correct solution

(1) correctly attempts to eliminate t from displacement equations

(2) correct solution

(must specify $\tan \alpha = \frac{5}{2}$ and $\tan \beta = \frac{3}{2}$ as $\alpha > \beta$)

(1) quadratic in $\tan \alpha$

(2) correct answer

(1) calculates time for one of the missiles

4 iii) When $x = 80, y = 55$

$$55 = 80 \tan \alpha - 6400 \frac{\sec^2 \alpha}{320}$$

$$55 = 80 \tan \alpha - 20(\tan^2 \alpha + 1)$$

$$20 \tan^2 \alpha - 80 \tan \alpha + 75 = 0$$

$$4 \tan^2 \alpha - 16 \tan \alpha + 15 = 0$$

$$(2 \tan \alpha - 3)(2 \tan \alpha - 5) = 0$$

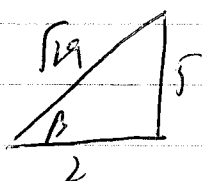
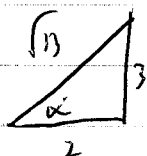
$$\tan \alpha = \frac{3}{2} \quad \tan \alpha = \frac{5}{2}$$

$$\therefore \tan \alpha = \frac{3}{2} \text{ \& \; } \tan \beta = \frac{5}{2}$$

② correct solution

① solves quadratic in $\tan \alpha$

iv)



$$x = 40 \cos \alpha$$

$$80 = 40 \cos \frac{2}{\sqrt{13}}$$

$$t = \sqrt{13}$$

$$\therefore \text{difference} = \sqrt{29} - \sqrt{13}$$

$$= 1.7796 \dots \text{secs}$$

$$x = 40 \cos \beta$$

$$80 = 40 \cos \frac{2}{\sqrt{29}}$$

$$t = \sqrt{29}$$

② correct answer

① calculates x

① calculates time for 1 missile

$$\begin{aligned}
 4b) i) \int_0^1 (1-x)^n dx &= \int_0^1 \left(\binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 + \dots + (-1)^n \binom{n}{n}x^n \right) dx \\
 &= \left[\binom{n}{0}x - \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} + \dots + (-1)^n \binom{n}{n}\frac{x^{n+1}}{n+1} \right]_0^1 \\
 &= \left(\binom{n}{0} - \binom{n}{1}\frac{1}{2} + \binom{n}{2}\frac{1}{3} + \dots + (-1)^n \binom{n}{n}\frac{1}{n+1} \right) - (0 - 0 + 0 + \dots + 0) \\
 &= \binom{n}{0} - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots + \frac{(-1)^n}{n+1}\binom{n}{n} \quad \textcircled{2} \text{ correct solution} \\
 &\quad \textcircled{1} \text{ Integrates RHS}
 \end{aligned}$$

$$\begin{aligned}
 b(ii) \int_0^1 (1-x)^{2018} dx &= \left[-\frac{(1-x)^{2019}}{2019} \right]_0^1 \\
 &= 0 + \frac{1}{2019}
 \end{aligned}$$

\textcircled{2} correct solution
 \textcircled{1} integrates $\int_0^1 (1-x)^n dx$

$$\begin{aligned}
 \therefore \text{from (i)} \quad \binom{2018}{0} - \frac{1}{2}\binom{2018}{1} + \frac{1}{3}\binom{2018}{2} + \dots + \frac{(-1)^{2018}}{2019}\binom{2018}{2018} &= \sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r} \\
 \therefore \sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r} &= \frac{1}{2019}
 \end{aligned}$$